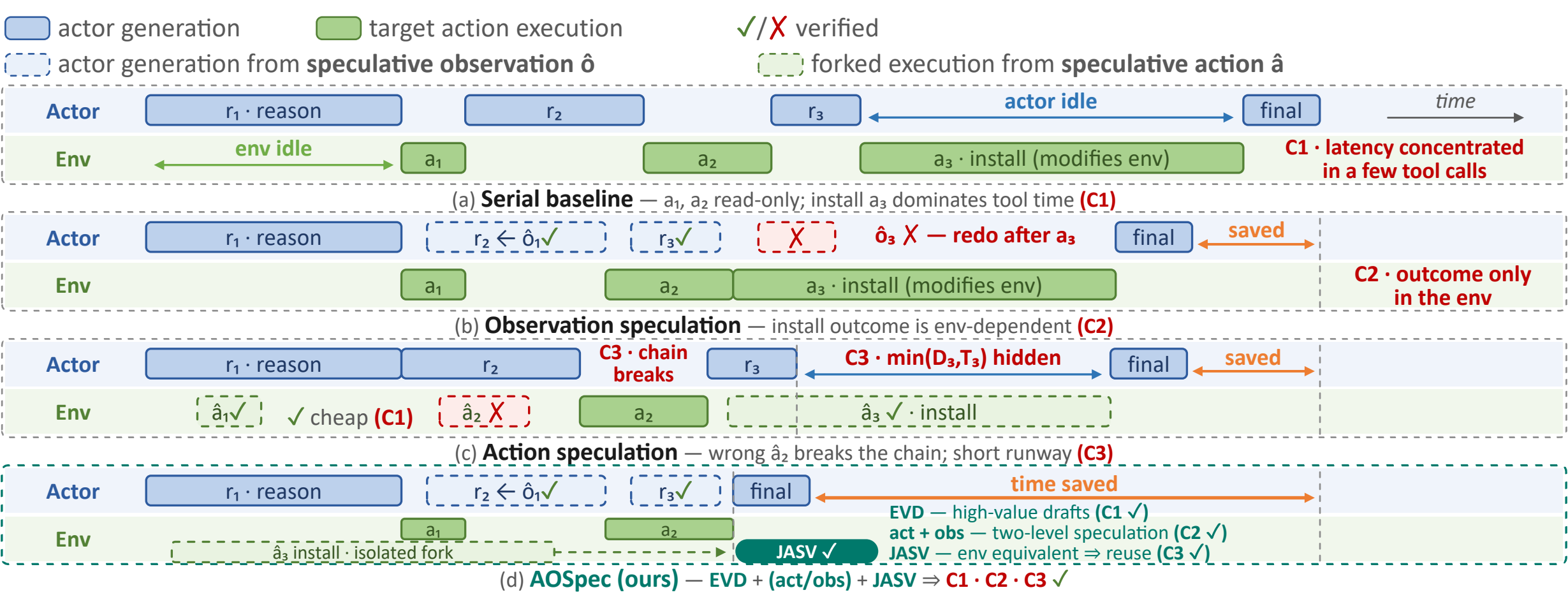

actor generation
target action execution
✓/✗ verified
actor generation from speculative observation ô
forked execution from speculative action â
Actor
r₁ · reason
r₂
r₃
actor idle
final
time
Env
env idle
a₁
a₂
a₃ · install (modifies env)
C1 · latency concentrated in a few tool calls
(a) Serial baseline — a₁, a₂ read-only; install a₃ dominates tool time (C1)
Actor
r₁ · reason
r₂ ← ô₁✓
r₃✓
✗
ô₃ ✗ — redo after a₃
final
saved
Env
a₁
a₂
a₃ · install (modifies env)
C2 · outcome only in the env
(b) Observation speculation — install outcome is env-dependent (C2)
Actor
r₁ · reason
r₂
C3 · chain breaks
r₃
C3 · min(D₃,T₃) hidden
final
saved
Env
â₁✓
✓ cheap (C1)
â₂ ✗
a₂
â₃ ✓ · install
(c) Action speculation — wrong â₂ breaks the chain; short runway (C3)
Actor
r₁ · reason
r₂ ← ô₁✓
r₃✓
final
time saved
Env
a₁
a₂
â₃ install · isolated fork
JASV ✓
EVD — high-value drafts (C1 ✓)
act + obs — two-level speculation (C2 ✓)
JASV — env equivalent ⇒ reuse (C3 ✓)
(d) AOSpec (ours) — EVD + (act/obs) + JASV ⇒ C1 · C2 · C3 ✓